BCA40204 – Data Warehousing and Data Mining

Introduction to Data Mining

MODULE 1

Table of contents

Sl. No.	Title / Contents	Page No
	Introduction to Data Mining	
1	Definition of Data Mining	1-1
1.1	Importance and Need of Data Mining	2-3
1.2	Applications of Data Mining	2-3
1.3	Data Mining vs Related Fields	3-4
1.4	A real life example of the data mining process	3-4
1.5	Knowledge Discovery in Databases (KDD)	4-6
1.6	Pattern evaluation	7-7
1.7	Knowledge Representation	7-8
1.8	Data Mining Process Models	9-10
1.9	Data Preprocessing	10-10
1.10	Exploratory Data Analysis (EDA)	11-15
1.11	Data Visualization Techniques	16-24
1.12	Key points to remember	25-26
	Practice Questions	26-29

1. What is Data Mining?

Data Mining is the process of discovering hidden patterns, useful knowledge, correlations, trends, and anomalies from large volumes of data stored in databases, data warehouses, or repositories.
The Goal: To turn raw data into gold (actionable insights).
The Flow: Data → Information → Knowledge → Decision.

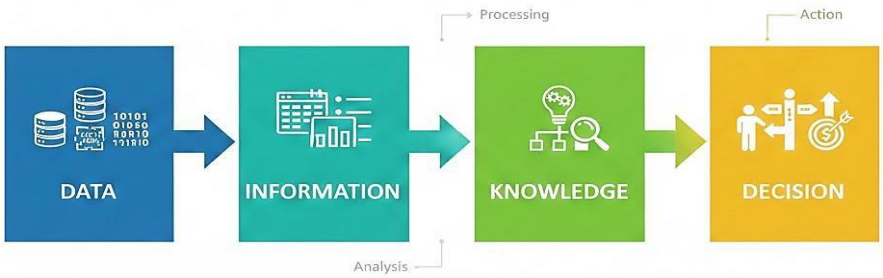

Figure 1. The Flow of data minning

1. **Data (The Input):** Raw, unorganized facts or symbols (e.g., numbers like $100, 150$). It lacks context and meaning on its own.
2. **Information (The Context):** Data that has been **analyzed** and organized to answer "Who, What, Where, When" (e.g., "$100 is Monday's sales").
3. **Knowledge (The Insight):** Information that has been **processed** to identify patterns and answer "How" (e.g., "Sales peak on Mondays due to promotions").
4. **Decision (The Action):** The application of knowledge to form a **strategy** and answer "What next?" (e.g., "Increase staffing on Mondays to handle the peak").

Stage	***Focus***	***Key Output***
Data	*Raw Facts*	*Input*
Information	*Context*	*Meaning*
Knowledge	*Patterns*	*Insight*
Decision	*Strategy*	***Action***

Table 1: Table format of the flow

1.1 Why Data Mining is Needed?

Data mining has become essential because we are currently "data rich but information poor." In an era of **Big Data**, we generate zettabytes of information that act as "noise," making data mining necessary to **filter** out meaningful patterns. Since modern datasets contain millions of rows and hundreds of variables, they far **surpass human cognitive limits**; where manual analysis would take months, algorithms provide the speed and multi-dimensional insight required to stay competitive. This shift transforms organizations from being **reactive to proactive**, moving beyond simple historical reporting to **predicting future trends**, such as customer churn or market shifts. Ultimately, it replaces subjective "gut feelings" with **objective, mathematical evidence**, enabling smarter decisions and real-time automation in critical areas like fraud detection and healthcare.

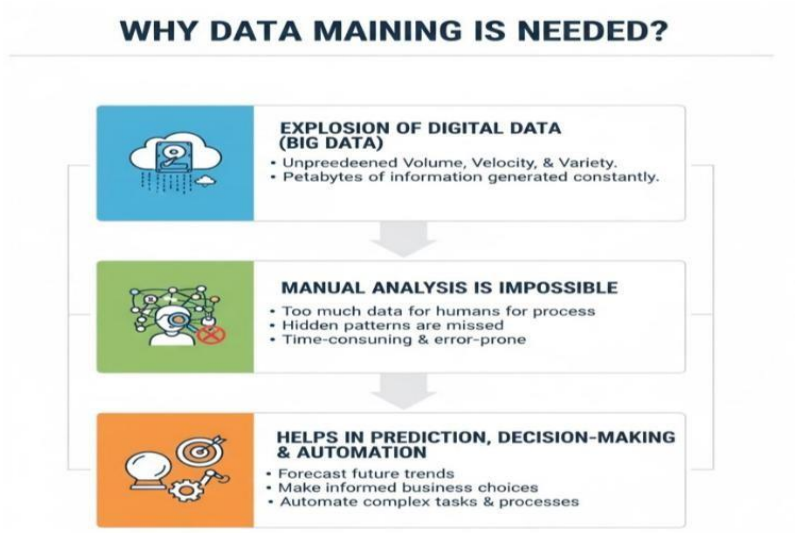

Figure 2: Importance of data minning

Core Benefits At a Glance

Challenge	*Data Mining Solution*
Data Overload	Information Extraction
Complexity	Pattern Recognition
Human Bias	Data-Driven Logic
Lag Time	Real-time Automation

Table 2

1.2 Applications of Data Mining

1. **Retail:** Maximizes revenue through **Market Basket Analysis** to find product associations and **Customer Segmentation** to target specific shopper groups.
2. **Banking:** Enhances security via **Fraud Detection** and minimizes financial risk through automated **Credit Scoring**.
3. **Healthcare:** Improves patient outcomes by using **Disease Prediction** models and assisting in faster, more accurate **Medical Diagnosis**.
4. **Education:** Boosts academic success by conducting **Student Performance Analysis** to identify and support at-risk learners early.
5. **Telecom:** Increases customer retention by using **Churn Prediction** to identify users likely to switch providers.
6. **Web & Social Media:** Drives user engagement through sophisticated **Recommendation Systems** that personalize content and advertisements.

1.3 Data Mining vs Related Fields

"While **Database Systems** give us the data and **Statistics/Machine Learning** provide the mathematical tools, **Data Mining** is the process of applying those tools to discover previously unknown, valid, and actionable information."

Field	***Primary Focus***	***Analogy***
Database Systems	**Storage & Retrieval:** Efficiently keeping and finding data.	The **Library:** Keeping books organized so you can find them.
Statistics	**Quantifying Uncertainty:** Testing hypotheses on data samples.	The **Audit:** Proving if a specific claim is true or false.
Machine Learning	**Prediction:** Teaching algorithms to learn and improve.	The **Student:** Learning from past exams to predict future answers.
Data Mining	**Discovery:** Finding hidden patterns in massive datasets.	The **Miner:** Digging through dirt to find unexpected gold.

Table 3

Key Differentiators

1. **Storage vs. Analysis:** Database systems *hold* the data; Data Mining *explores* it.
2. **Proving vs. Finding:** Statistics *tests* a known idea; Data Mining *finds* a new one.
3. **Accuracy vs. Understanding:** Machine Learning wants the *right answer*; Data Mining wants the *reason why*.

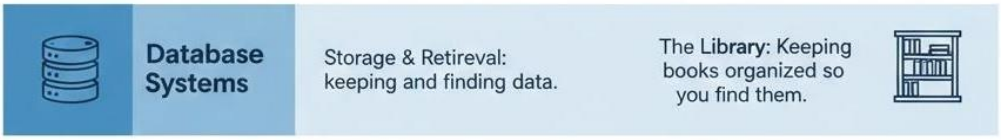

Figure 3: Database

Figure 4: Statistics

Figure 5 : Machine Learning

Figure 6: Data mining

1.4 A real life example of the data mining process

In healthcare, data mining transforms patient history and clinical data into life-saving insights. Below is a detailed look at how this works in practice, specifically for **Chronic Kidney Disease (CKD)** prediction.

1. The Context: Predictive Diagnosis

Medical diagnosis often relies on identifying patterns across dozens of lab results (blood pressure, glucose, creatinine, etc.). Data mining algorithms can analyze these variables simultaneously to identify the early onset of a disease before symptoms become severe.**Example Case:** A hospital uses a "Random Forest" model to analyze patient blood work. The model identifies that a specific combination of high blood pressure and specific gravity in urine indicates a 90% probability of early-stage kidney disease, allowing doctors to intervene months earlier than usual.

2. Data Source

For this example, we use the **UCI Machine Learning Repository: Chronic Kidney Disease Dataset**. It contains 400 instances with 24 attributes (like age, blood pressure, and specific gravity).

3. Implementation Example (Python)

This code demonstrates a simplified version of the **Data Mining step** in the KDD process: training a model to classify whether a patient is healthy or ill.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
```

STEP 1 & 3: DATA SELECTION & INTEGRATION

(Imagine merging multiple hospital files here)

```
data = pd.read_csv('kidney_disease.csv')
```

STEP 2: DATA CLEANING

(Removing 'noise' and handling missing entries)

```
data.dropna(inplace=True)
```

STEP 4: DATA TRANSFORMATION

(Feature selection (X) and Labeling (y))

```
X = data[['bp', 'gl', 'al']]
y = data['classification'].map({'ckd': 1, 'notckd': 0})
```

(Splitting data (Preparation for Mining))

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

STEP 5: DATA MINING
(The core algorithm extracting patterns)

```
model = RandomForestClassifier(n_estimators=100)
model.fit(X_train, y_train)
```

STEP 6: PATTERN EVALUATION
(Assessing if the discovered patterns are accurate)

```
predictions = model.predict(X_test)
print(f"Discovery Accuracy: {accuracy_score(y_test, predictions) * 100:.2f}%")
```

STEP 7: KNOWLEDGE REPRESENTATION
(Visualizing the most important factors for the doctor)

```
feat_importances = pd.Series(model.feature_importances_, index=X.columns)
feat_importances.nlargest(3).plot(kind='barh')
plt.title("Most Important Factors for Kidney Disease")
plt.show()
```

1.5 Knowledge Discovery in Databases (KDD)

KDD is a **complete process**, Data Mining is only one step.
The scientific approach to turning raw bits into knowledge.

1. **Data Selection:** Picking relevant data (e.g., specific medical lab results).
2. **Data Cleaning:** Removing noise, duplicates, and handling missing values.
3. **Data Integration:** Combining data from different sources(e.g., merging Lab).
4. **Data Transformation:** Normalizing data (scaling) or converting types.
5. **Data Mining:** Applying algorithms (Random Forest, Clustering) to find patterns.
6. **Pattern Evaluation:** Interpreting if the found pattern is valid and useful.
7. **Knowledge Representation:** Presenting findings via dashboards and visualization.

1. Data Selection

Mathematical Concept:
Based on **Set Theory**. It involves defining a subset **S** from a universal database **D** such that **S ⊆ D**, where elements are chosen based on specific relevance criteria.
Statistical Application: Uses **Feature Selection** techniques.
Information Gain: Measures the reduction in entropy.
Variance (σ^2): If a variable has **Zero Variance ($\sigma^2 = 0$)**, it contains no discriminative power and is mathematically discarded.
Formula for Variance: $\sigma^2 = (1 / n) \times \Sigma (x_i - \mu)^2$

2. Data Cleaning

Mathematical Concept: Relies on **Probability Theory** and **Descriptive Statistics** to handle "noisy" data.

Imputation: Replace missing values with Mean (μ), Median, or Mode.
Z-score Formula: $z = (x - \mu) / \sigma$

***Note: If $|z| > 3 \rightarrow$ the data point is an outlier (beyond 99.7% of normal distribution).*

4. Data Integration

Mathematical Concept: Utilises **Relational Algebra** (Joins and Unions) to merge disparate datasets using primary/foreign keys.
Pearson's Correlation Coefficient:
$r = \Sigma[(x_i - \bar{x})(y_i - \bar{y})] / \sqrt{[\Sigma(x_i - \bar{x})^2 * \Sigma(y_i - \bar{y})^2]}$
If **r = 1.0**, variables are perfectly correlated (redundant). Removing one prevents **Multi-collinearity**, which can destabilize mathematical models.

5. Data Transformation

Mathematical Concept: Uses **Linear Algebra** to change the scale or functional form of the data.
Min–Max Scaling:
$x_new = (x - x_{_min}) / (x_{_max} - x_{_min})$
Z-score Normalisation: Re-centers data to have a mean .
$x_new = (x - \mu) / \sigma$.

6. Data Mining

Mathematical Concept: The application of Optimization Algorithms and Distance Metrics.
Classification: Uses **Gini Impurity** to determine optimal splits in decision trees:
$Gini = 1 - \Sigma(p_i)^2$
Clustering: Uses **Euclidean Distance** to calculate the proximity between data points p & q:
Euclidean Distance (for Clustering): $d(p, q) = \sqrt{\Sigma(p_i - q_i)^2}$

1.6 Pattern Evaluation

Mathematical Concept: Employs **Threshold Logic** to determine the validity of discovered patterns.
Accuracy:
(TP + TN) / Total
Precision:
TP / (TP + FP)
Recall:
TP / (TP + FN)
Support (Association Rule):
(Number of transactions containing A and B) / (Total transactions)
Confidence (Association Rule):
(Support of A and B) / (Support of A)

1.7 Knowledge Representation

Principal Component Analysis (PCA):
Let Σ be the covariance matrix of data. Find Eigenvalues (λ) and Eigenvectors (v) such that: $\Sigma v = \lambda v$
Project data onto new axes using: $Z = X \times v$
**Goal: Reduce dimensions while retaining maximum variance.

Core Takeaway

Aspect	*Definition*
Input	Raw, messy, disconnected data.
Middle	Mathematical cleaning, scaling, and algorithmic "mining."
Output	Actionable, visualized knowledge.

Table 4

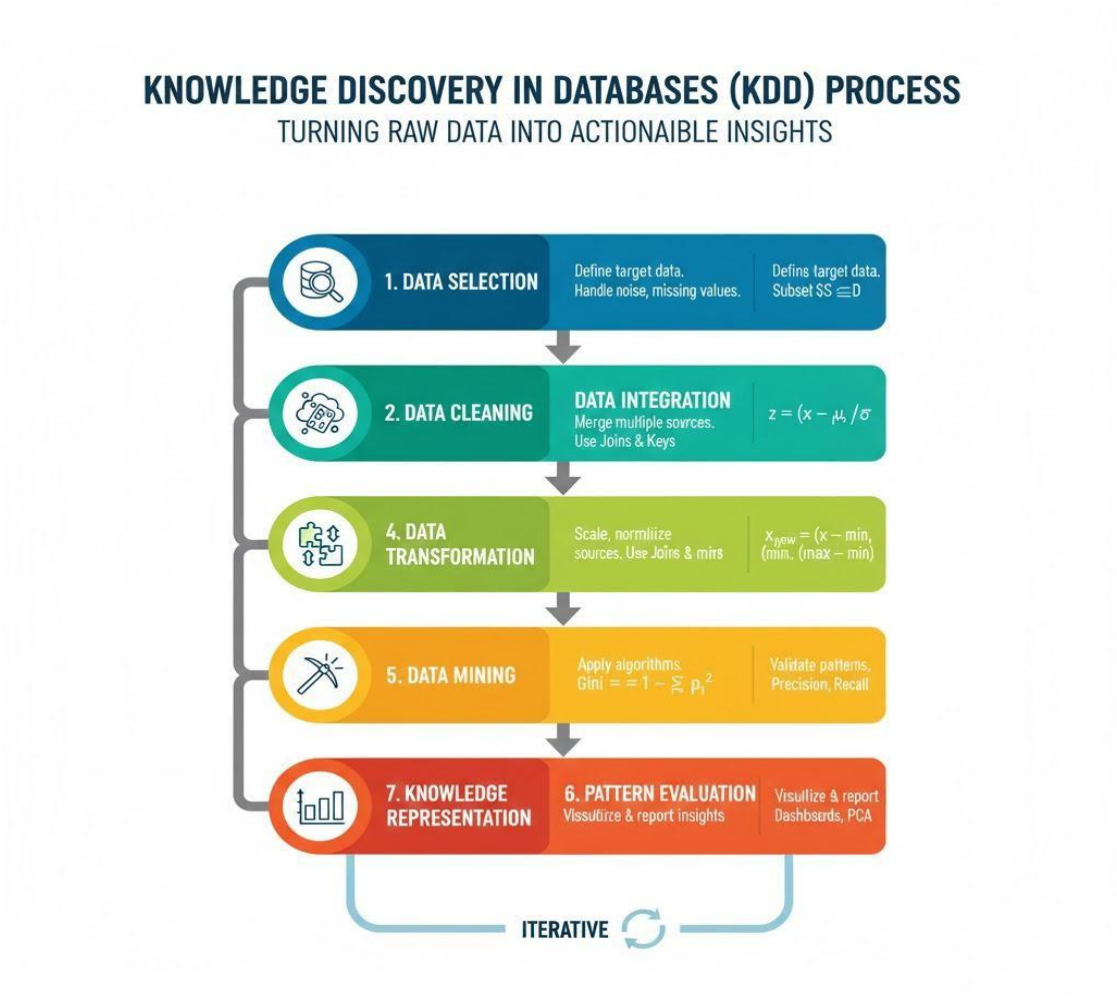

Figure 7

1.8 Data Mining Process Models

While KDD is a technical, scientific framework, **CRISP-DM** is the **industry-standard business process** for data mining. It acknowledges that data mining isn't just about math; it's about solving a specific problem for a company or organization.

1. Business Understanding

Define what the business actually needs. We don't look at data yet. Instead, we ask: "What is the problem?"**Example:** A telecom company wants to know: *"Why are customers leaving for our competitors, and how can we stop them?"*

2. Data Understanding

Explore the available data to see if it can solve the business problem. Collect initial data, check for quality issues, and identify interesting subsets. **Example:** Looking at call logs, monthly bills, and customer service complaints to see if there is a link between "dropped calls" and "leaving the company."

3. Data Preparation (The 80% Rule)

Construct the final dataset for the modeling tool. This aligns with KDD Steps 2, 3, and 4 (Cleaning, Integration, and Transformation). We select tables, clean the "noise," and scale the values.In the industry, this step usually takes **80% of the project time**.

4. Modeling

Apply various mathematical algorithms to the data.We select and apply techniques like Random Forest, Neural Networks, or K-Means. This is the "Data Mining" step of KDD.**Example:** Building a model that assigns a "Churn Risk Score" to every customer.

5. Evaluation

Check if the model actually meets the **Business Objectives** defined in Step 1.We don't just check for "Accuracy"; you check for "Value."**Example:** The model is 99% accurate, but it only identifies customers *after* they have already left. This is a **technical success** but a **business failure**. You must go back to Step 1 or 2.

6. Deployment

Put the knowledge into action. Integrate the model into the company's daily operations. **Example:** The "Churn Risk Score" is sent to the customer service dashboard. When a high-risk customer calls, the agent sees a pop-up offering a 20% discount to keep them.

CRISP-DM vs. KDD: The Key Difference

Feature	*KDD*	*CRISP-DM*
Origin	Academic / Scientific	Industry / Business
Focus	How to find patterns accurately.	How to solve a business problem profitably.
Structure	Linear (Step 1 to 7)	**Cyclical** (It never truly ends).

Table 5

1.9 Data Preprocessing

Data Preprocessing is the process of transforming raw, "dirty" data into a clean, organized format suitable for data mining. In the real world, data is often **incomplete** (missing values), **noisy** (errors/outliers), and **inconsistent** (different formats).Technically, **KDD is not just data preprocessing**, but **Data Preprocessing makes up the most critical part of KDD.**KDD is the **entire journey** from raw data to a final decision. Preprocessing is the **preparation phase** that takes place in the middle of that journey.In the 7-step KDD process, **Steps 2, 3, and 4** are collectively known as **Data Preprocessing**.
Data Preprocessing is a subset of KDD.**KDD** is the **lifecycle** that contains Preprocessing.If you skip Preprocessing, the Data Mining step of KDD will produce "Garbage."In real-world data science projects, **80% of the work** happens in these preprocessing steps.

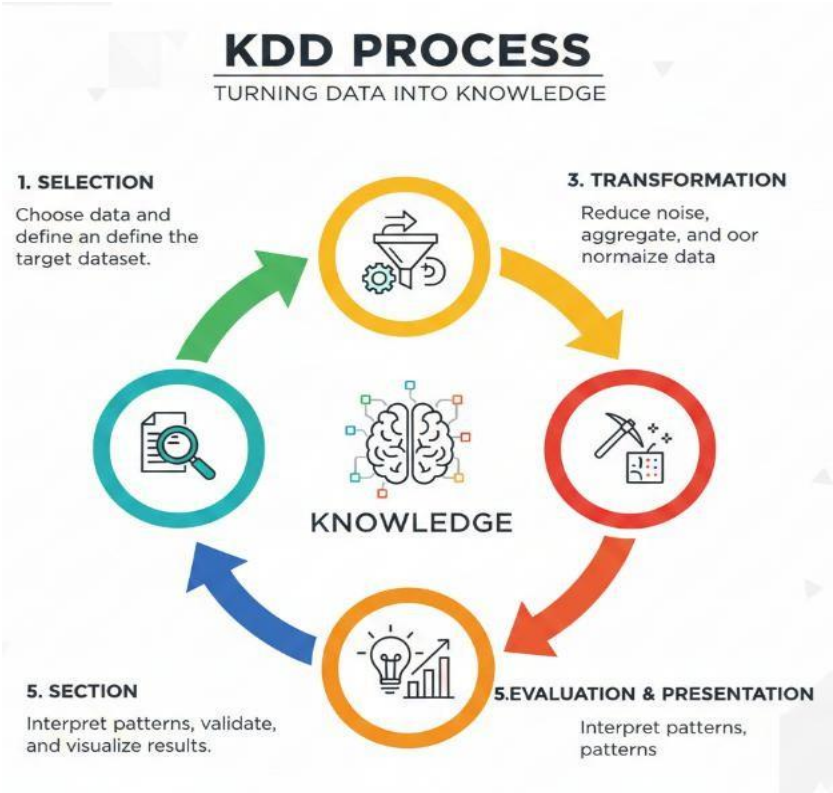

Figure 8

1. **KDD (The Whole Factory):** Includes the business goal, the cleaning, the actual machine learning (Data Mining), and the final charts.
2. **Preprocessing (The Filter):** This is just the "preparation" of the data so the Data Mining algorithm doesn't fail.

KDD Step	*Preprocessing Task*	*Purpose*
Data Cleaning	**Cleaning**	Fixes errors, removes outliers, and fills in missing gaps.
Data Integration	**Integration**	Merges different databases and removes redundant info.
Data Transformation	**Transformation**	Scales numbers (Normalization) so the math is "fair."

Table 6

1.10 Exploratory Data Analysis (EDA)

EDA helps understand data **before modeling**. **Exploratory Data Analysis (EDA)** is the process of **understanding, summarizing, and visualizing data before applying any modeling or machine learning algorithms**. Its main goal is to **discover patterns, detect anomalies, test assumptions, and check data quality**.

Why EDA is Important ?

1. Understand the **structure and nature of data.**
2. Identify **missing values** and **outliers.**
3. Detect **relationships between variables.**
4. Decide **feature selection and transformation.**
5. Prevent **wrong model assumptions.**

EDA ensures **data quality and reliability** before modeling.EDA ensures **data quality and reliability** before modeling.Descriptive statistics in EDA summarize **central tendency and dispersion**. EDA helps identify **missing values and outliers** in the dataset.EDA reveals **relationships and correlations** between variables.EDA prevents **wrong model assumptions**.

Key Steps in the EDA

Step No.	EDA Step	Purpose / Description	Common Tools / Methods
1	Data Understanding	Identify number of rows, columns, and data types	`df.shape`, `df.info()`
2	Descriptive Statistics	Summarize central tendency and spread of data	`df.describe()`
3	Missing Value Analysis	Detect and quantify missing data	`df.isnull().sum()`
4	Outlier Detection	Identify extreme or abnormal values	Box plot, IQR, Z-score
5	Data Distribution Analysis	Understand spread and skewness of variables	Histogram, KDE plot
6	Relationship Analysis	Find correlation and dependency between variables	`df.corr()`, scatter plot
7	Categorical Data Analysis	Analyze frequency of categories	Bar chart, `value_counts()`
8	Data Visualization	Visual inspection of patterns and trends	Matplotlib, Seaborn
9	Feature Engineering Decisions	Decide transformations and feature selection	Scaling, encoding
10	Data Quality Check	Ensure consistency and correctness	Duplicates, invalid values

Table 7

In data science, the "detective phase" is where you investigate a dataset to understand its internal structure, detect anomalies, and test assumptions before applying complex machine learning models. Techniques that have been used in the following:

1. Summary Statistics (Mean, Variance)

Summary statistics provide a high-level numerical snapshot of your data. Instead of looking at thousands of individual rows, you use a few numbers to describe the whole group.

1.1. Measures of Central Tendency:

Mean: The average value. It's the "centre of gravity" for your data.
Median: The middle value. It is more robust than the mean because it isn't easily swayed by extreme outliers.

1.2. Measures of Spread (Dispersion):

Variance: Measures how far each number in the set is from the mean. High variance means the data is spread out; low variance means it's clustered together.
Standard Deviation: The square root of variance, providing a spread measure in the same units as the data itself.

Summary statistics provide a numerical snapshot of your data's "centre" (Mean) and its "spread" (Variance) to identify general patterns and outliers at a glance.

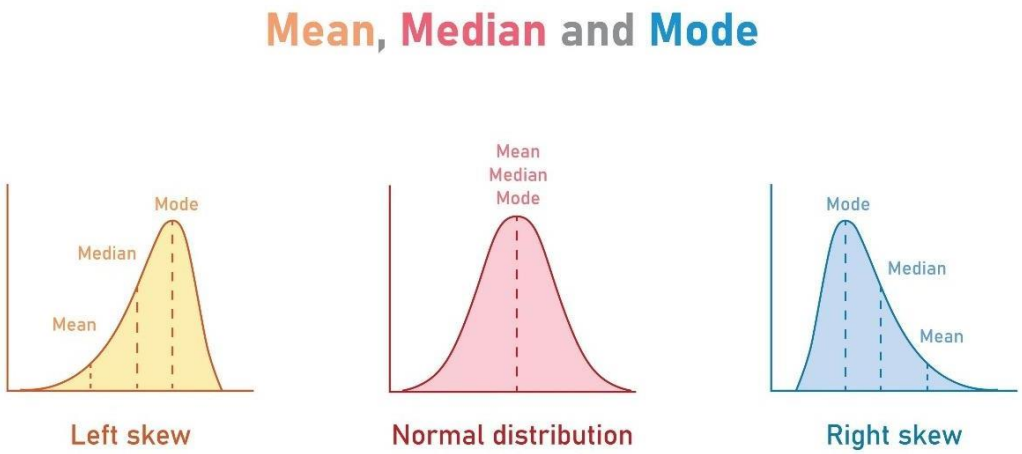

Figure 9

2. Data Distribution

This tells you the "shape" of your data. Understanding the distribution is vital because many machine learning algorithms (like Linear Regression) assume the data follows a specific shape, usually a **Normal Distribution**. **Histograms:** These show the frequency of data points. They help you see if your data is "skewed" (leaning to one side) or "bimodal" (having two peaks). **Box Plots:** These are the best tools for spotting **outliers**—data points that are so far from the rest that they might be errors or unique cases that need special attention.

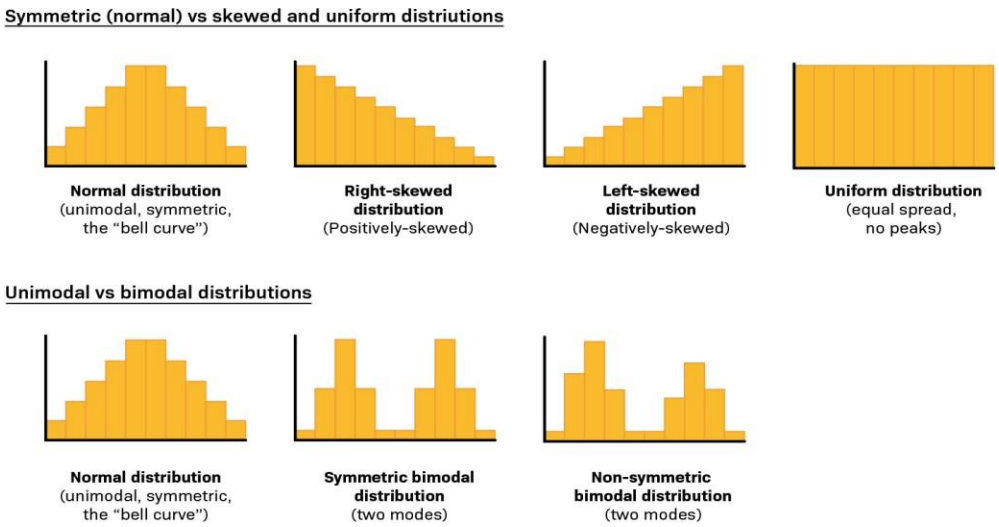

Figure 10

3. Correlation analysis

This examines the relationship between two variables. It helps you understand if one variable changes when another does, which is essential for **Feature Selection** (choosing which data matters most).
Positive Correlation: As X increases, Y increases (e.g., height and weight).
Negative Correlation: As X increases, Y decreases (e.g., altitude and temperature).
Correlation Coefficient (r): A value between -1 and +1.

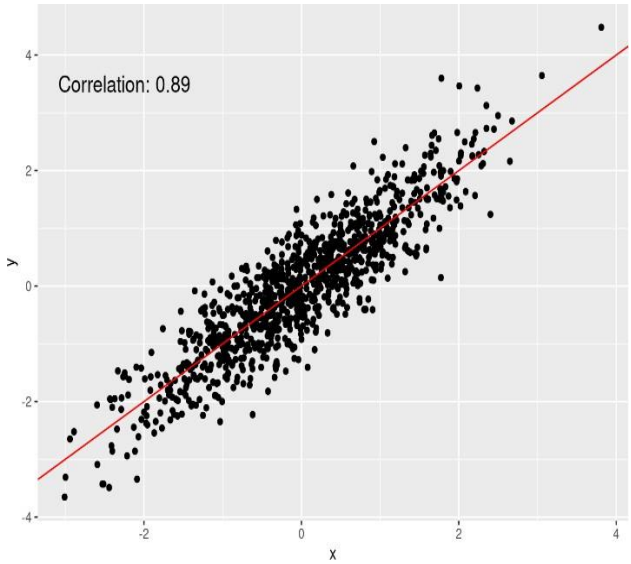

Figure 11 : If r = 1: Perfect positive relationship.

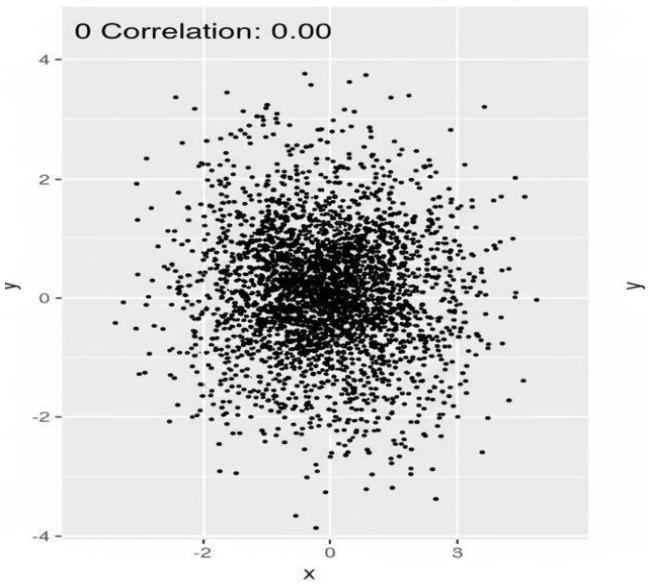

Figure 12 : r = 0: No linear relationship.

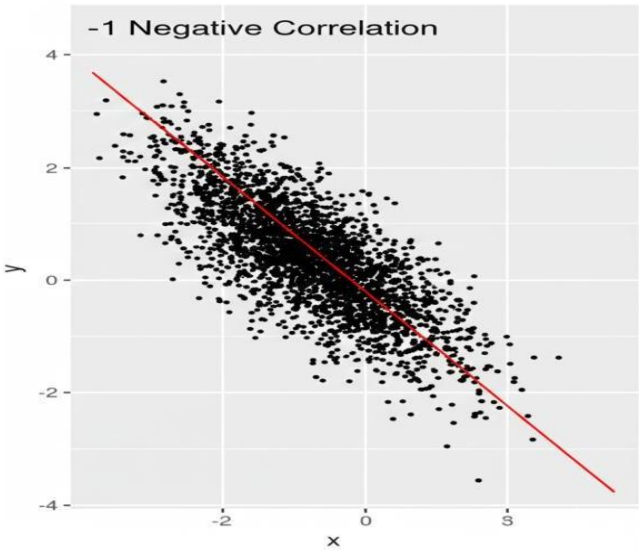

Figure 13: r = -1: Perfect negative relationship.

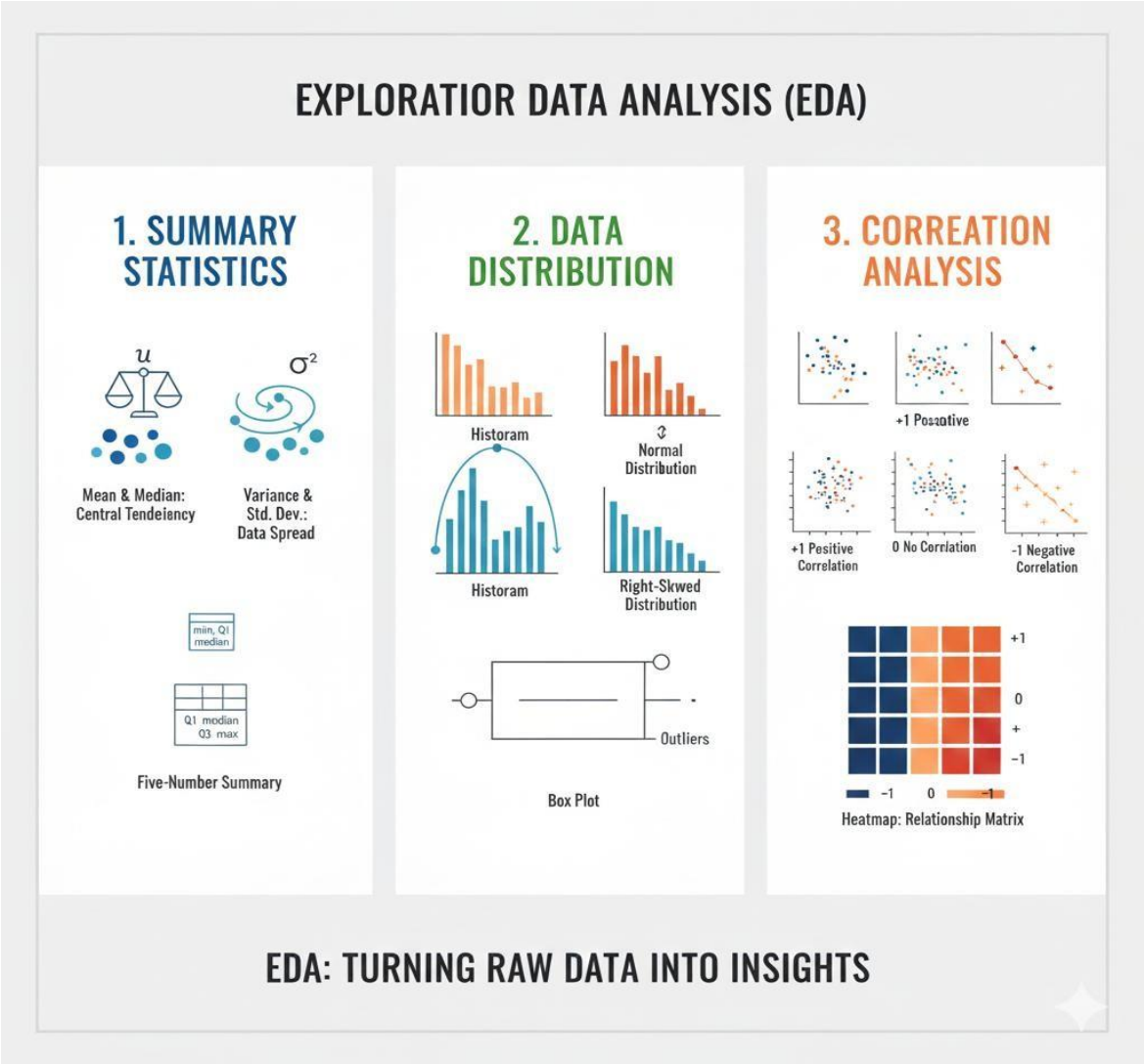

Figure 14 : Infographic of exploratior data analysis

1.11 Data Visualization Techniques

Data visualization is the process of translating complex data into graphical formats. This allows you to spot trends, outliers, and patterns that are nearly impossible to see in a raw spreadsheet.

1. **Histogram:**

   A histogram displays the **distribution of a single numerical variable** by grouping data into intervals called "bins."The Concept is that each bar's height represents the frequency (count) of data points within that bin. Best use **is t**o see if your data is "normal" (bell-shaped), skewed to one side, or contains gaps.It essentially converts continuous data into discrete blocks, making it easy to identify the "center" and "spread" of your data visually.

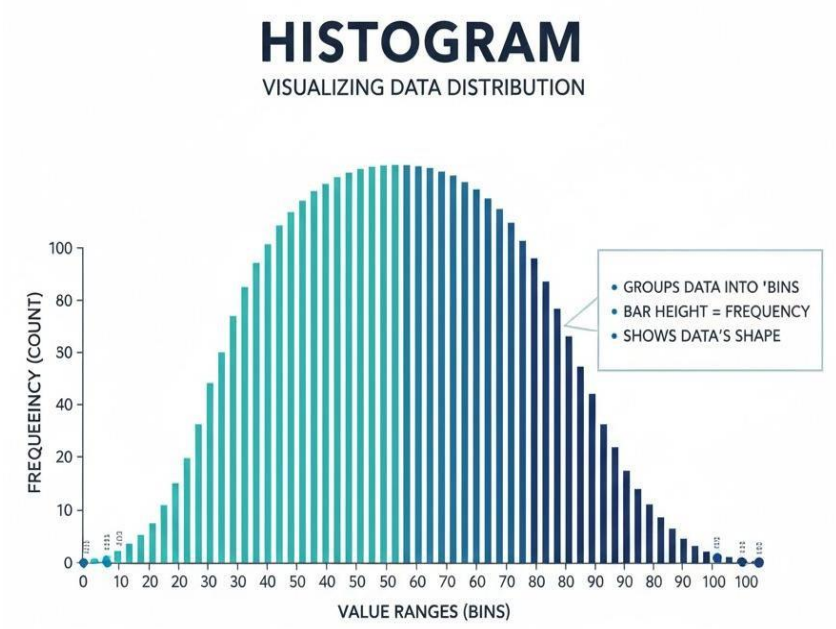

Figure 15

Figure 16

2. Box plot:

A **Box Plot** (also known as a **Box-and-Whisker Plot**) is a standardized way of displaying the distribution of data based on a **five-number summary**. It is one of the most effective tools for identifying outliers and comparing distributions between different groups.

The Anatomy of a Box Plot

A box plot is composed of several key parts, each representing a specific statistical milestone in data.
The Median (Q2): The line inside the box. It marks the middle of the dataset(50th percentile).
The Box (IQR): This represents the **Interquartile Range**, which contains the middle 50% of your data.
Lower Hinge (Q1): The bottom of the box (25th percentile).
Upper Hinge (Q3): The top of the box (75th percentile).
Whiskers: Lines extending from the box to the smallest and largest values that are *not* outliers.
Outliers: Individual dots or asterisks plotted beyond the whiskers.

How Outliers are Calculated (The 1.5 IQR Rule)

Data scientists use a specific formula to decide if a point is an outlier or just a "whisker":
Calculate IQR: IQR = Q3 - Q1
Lower Bound: Q1 - (1.5 \times IQR)
Upper Bound: Q3 + (1.5 \times IQR)

**Any data point falling outside these bounds is mathematically flagged as an outlier.

What the Shape Tells You

The symmetry of the box plot reveals the skewness of your data:
Symmetric: The median is in the center of the box, and the whiskers are roughly equal in length.
Right Skewed (Positive): The median is closer to the bottom of the box, and the top whisker is longer.
Left Skewed (Negative): The median is closer to the top of the box, and the bottom whisker is longer.

Skewness – Small Numerical Example

Consider the dataset:
Data:2, 3, 4, 5, 20
Step1. Find the mean

$$\text{Mean} = \frac{2+3+4+5+20}{5} = \frac{34}{5} = 6.8$$

Step2 : Find the median

$$\text{Median} = 4$$

Step 3: Skewness Interpretation

Mean > median indicates **Positive (Right) Skewness**.

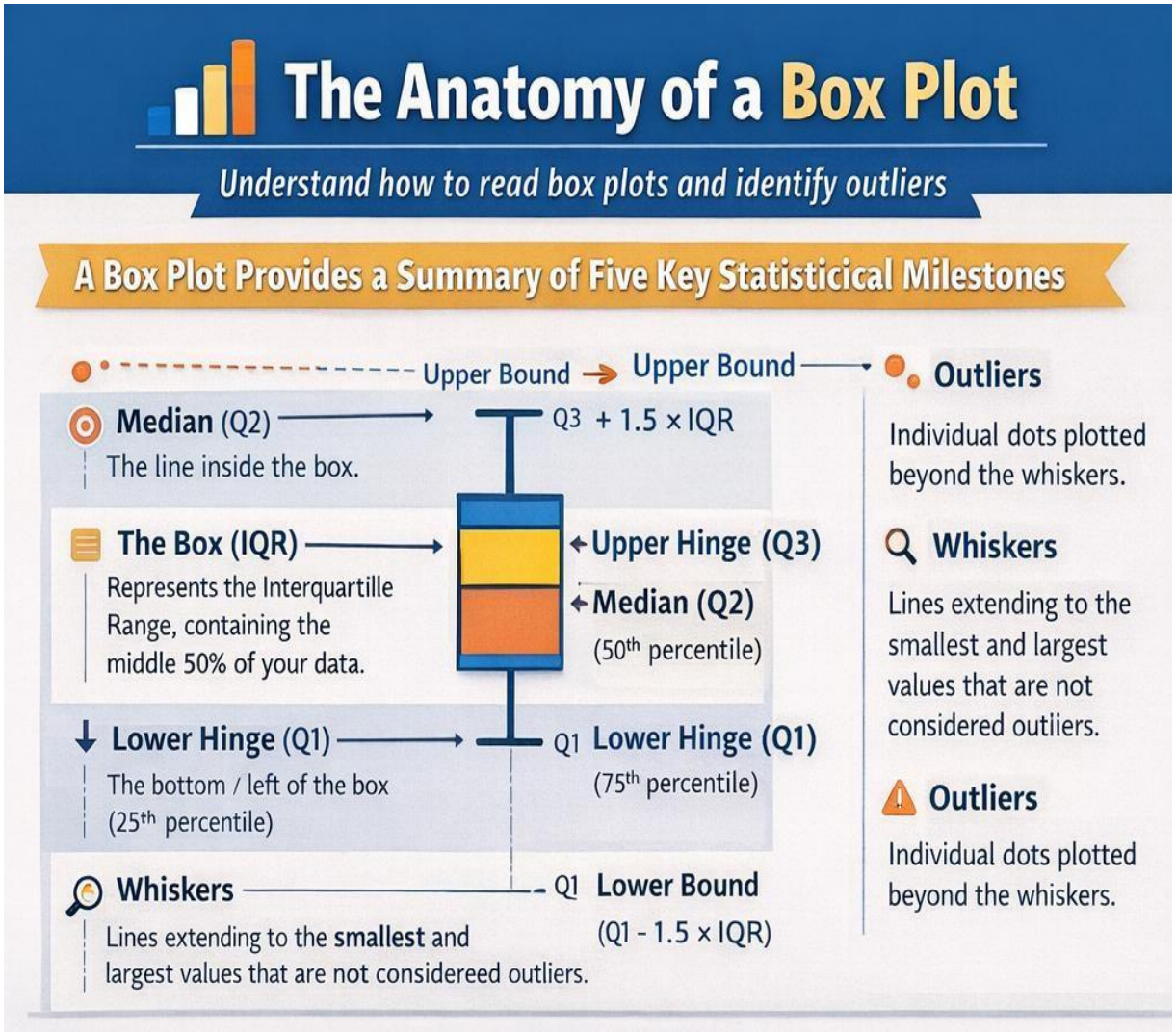

Figure 17

A box plot visually summarizes data using five statistics—minimum, Q1, median (Q2), Q3, and maximum—highlighting spread, skewness, and outliers via the IQR and whiskers.

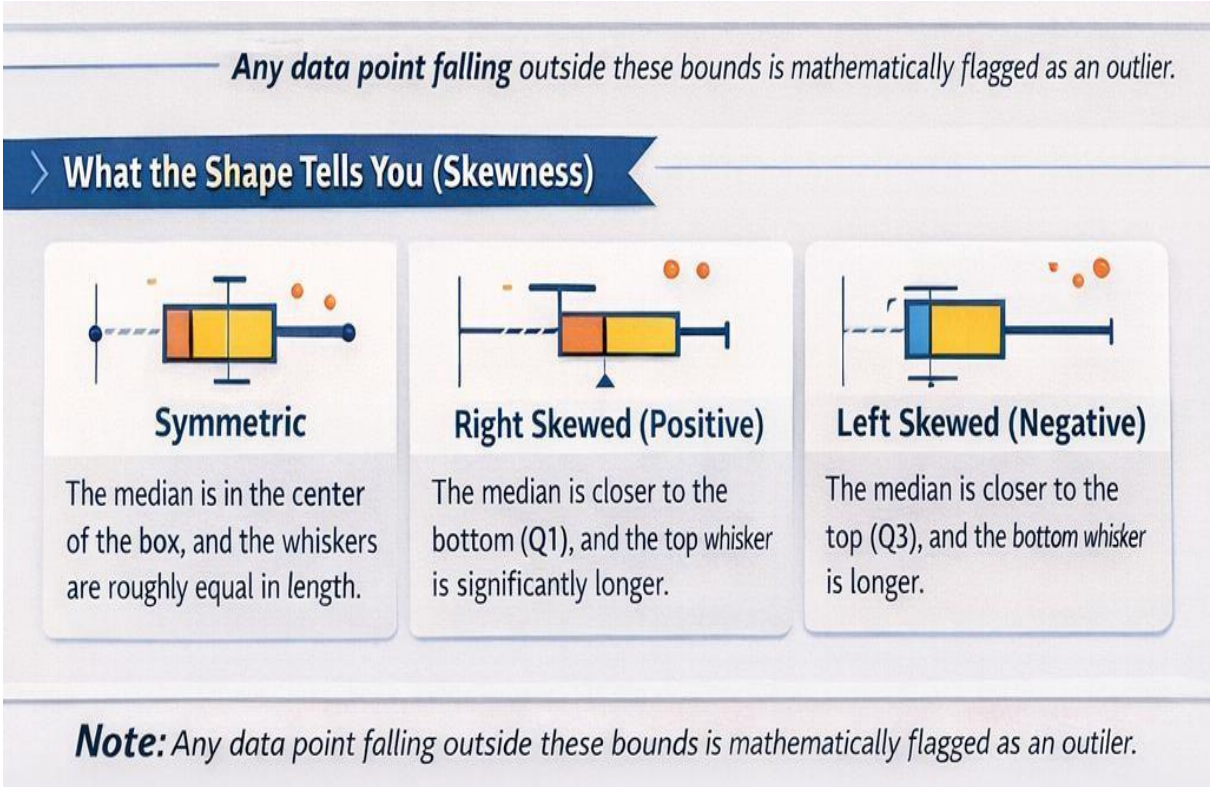

Figure 18

3. Scatter plot:

A **scatter plot** is a **two-dimensional graphical representation** used to **display the relationship between two numerical (quantitative) variables**. Each point on the graph represents **one observation**. A **scatter plot** shows how one variable changes with respect to another by plotting data points on an **X–Y coordinate system**. **Purpose** to **identify relationships, trends, patterns, clusters, and outliers** between variables.

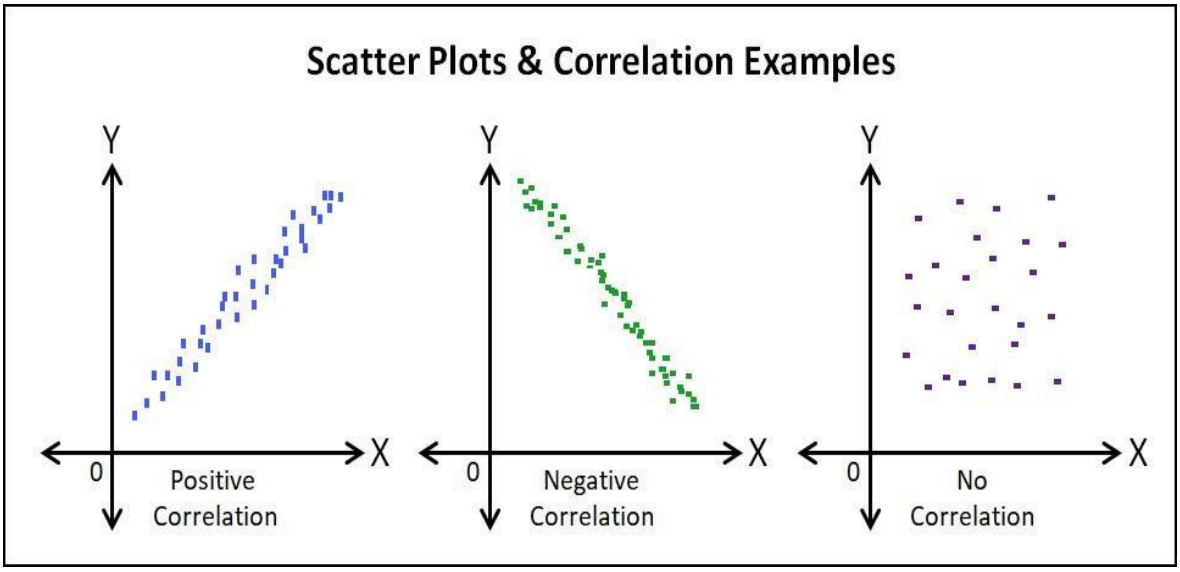

Figure 19

Line of Best Fit & The Strength of Correlation (R)

When analyzing a Scatter Plot It is often helpful to add what is called the Line of Best Fit in order to visualize the strength of your correlation.

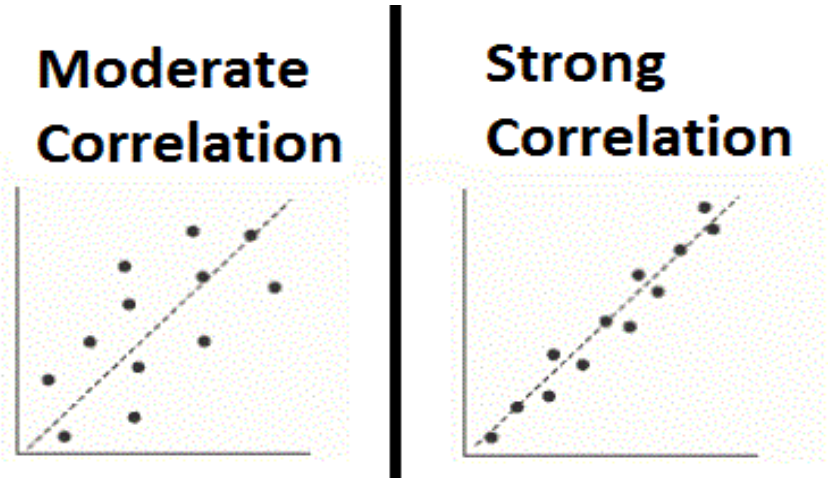

Figure 20

The R Value, which is also called **Pearson's Correlation Coefficient** is a measure of the strength and direction of the linear relationship between two variables. R Values range from +1 to -1, with +1 being a perfect positive correlation, -1 being a perfectly negative correlation and 0 being absolutely no measurable correlation.

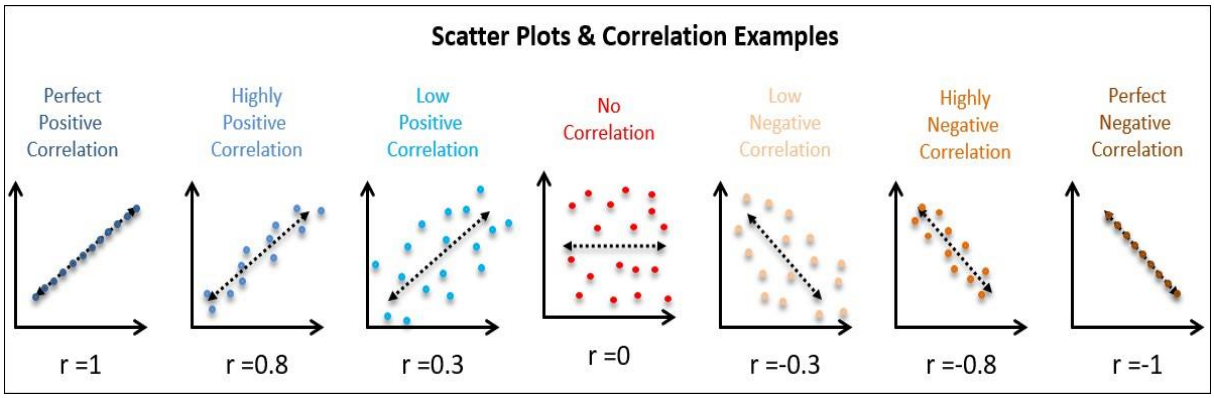

Figure 21

The Pearson correlation coefficient (R) is an indicator of the strength of a linear relationship between two variables, however its value generally does not completely characterize their relationship. Correlation does not equal causation. Just because a scatter plot shows a perfect upward trend doesn't mean X caused Y. For example, ice cream sales and shark attacks both go up in the summer (positive correlation), but eating ice cream does not cause shark attacks—the "hidden" variable is the warm weather.

The Difference between R & R^2

R = The Correlation Coefficient
R^2 = The Coefficient of Determination

R^2 represents the % of the total variation in *y*, that can be explained by the linear relationship between *x* & *y*.

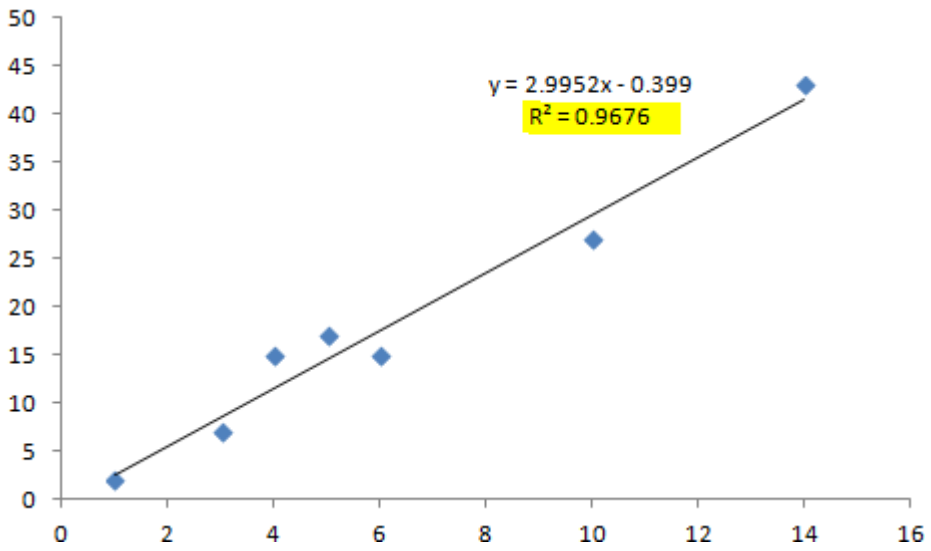

Figure 22

Let's say you measure r and it turns out to be 0.866. Then you calculate $R^2 = 0.75$ by just squaring your R value. You can then interpret your data to mean that 75% of the total variation in ***y*** can be explained by the linear relationship between ***x*** & ***y.***

4. **Heat map:**

A heat map is a two-dimensional data visualization technique that represents numerical values using colors. The intensity of color indicates the magnitude of the value.Often used for matrices, geographic areas, or user behavior (e.g., website clicks).

How It Works

1. **Grid/Area:** Data is mapped onto a grid or a spatial layout.
2. **Color Scale:** A gradient of colors (e.g., cool blue to warm red) is assigned to the range of data values.
3. **Visualization:** Each cell or point is colored according to its value, making magnitudes instantly visible.Example (Correlation Matrix):

	A	B	C
A	1.0	0.8	0.2
B	0.8	1.0	0.5
C	0.2	0.5	1.0

Why Use It?

1. **Pattern Detection:** Easily spot clusters, anomalies, and overall trends.
2. **Data Density:** Visualize large datasets in an intuitive, compact form.
3. **Insight Generation:** Quickly understand where attention or action is needed.

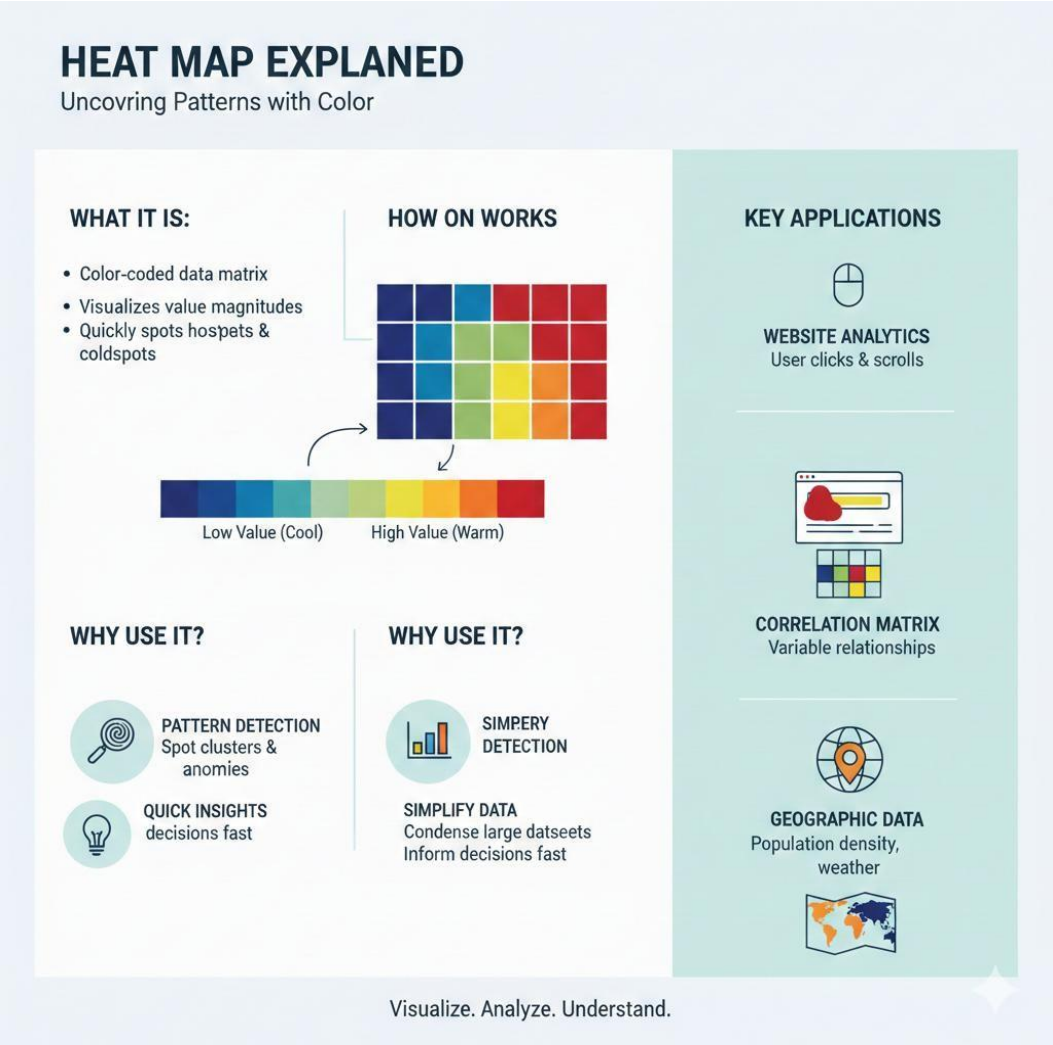

Figure 23

The darker or more intense the color, the higher the value; the lighter the color, the lower the value. A heat map emphasizes ***patterns and intensity***, not exact numeric values.

5. Bar chart:

A **bar chart** (also called a **bar graph**) is a **graphical representation of categorical data** using **rectangular bars**, where the **length or height of each bar** is **proportional to the value it represents.**

Structure of a Bar Chart

1. **X-axis:** Displays **categories** or qualitative data being compared.
2. **Y-axis:** Shows **numerical values** such as frequency, count, or amount.
3. **Bars:** Rectangular shapes whose **height/length represents the value** of each category.
4. **Width of bars:** Kept **equal** for all categories to ensure fair comparison.
5. **Gap between bars:** Indicates that the data is **discrete and categorical**.
6. **Scale:** Defines how numerical values are measured and represented on the axis.

Common Types of Bar Charts

Depending on how much data you have, you might use different styles:

1. **Simple Bar Chart:** Compares single values across categories. *Example: Revenue of 5 different stores.*
2. **Grouped (Clustered) Bar Chart:** Places bars side-by-side to compare sub-groups within a category. *Example: Revenue of 5 stores, split by "Online" vs. "In-Store" sales.*
3. **Stacked Bar Chart:** Places sub-groups on top of each other to show the **total** value while still seeing the individual parts. *Example: Total store revenue, with sections of the bar colored for different product types.*
4. **Horizontal Bar Chart:** Best when your category labels are very long (like full sentences or long names) because they are easier to read from left to right.

Bar Chart vs. Histogram (The "Touching" Rule)

People often confuse these two because they both use bars, but they serve different purposes:

Feature	*Bar Chart*	*Histogram*
Data Type	**Categorical** (Discrete names/groups)	**Continuous** (Number ranges/bins)
Gaps	There are **gaps** between bars.	Bars **touch** each other.
Example	Sales by Country	Number of people in age groups (10-20, 21-30)

Table 8

Types of bar chart

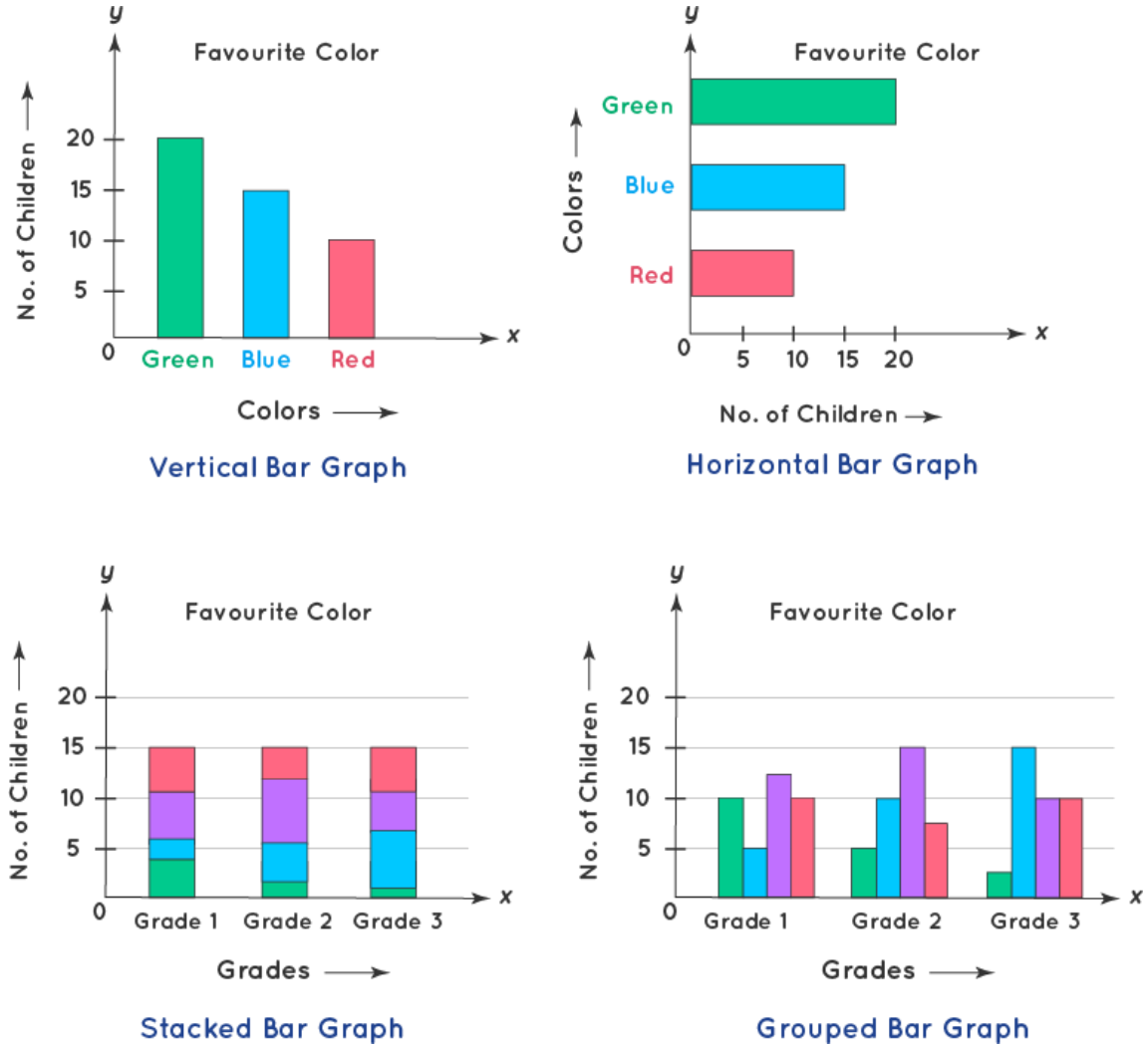

Figure 24

1.12 Key points to remember

1. **Data Mining** is the process of discovering hidden patterns and useful knowledge from large datasets.
2. The primary goal of data mining is to convert **raw data into actionable decisions**.
3. The fundamental flow is **Data → Information → Knowledge → Decision**.
4. Modern society is **data rich but information poor**.
5. Data mining enables **predictive, proactive, and automated decision-making**.
6. It replaces human intuition with **objective, data-driven logic**.
7. **KDD** is the complete knowledge discovery process; **data mining is only one step**.
8. Data preprocessing (cleaning, integration, transformation) consumes about **80% of project effort**.
9. Poor preprocessing results in **garbage output**.
10. **CRISP-DM** is the industry-standard, business-oriented data mining model.
11. Classification commonly uses **Gini Index** to split data.
12. Clustering measures similarity using **Euclidean distance**.
13. Correlation coefficient (**r**) measures strength and direction of linear relationships.
14. **Correlation does not imply causation**.
15. Model performance is evaluated using **accuracy, precision, and recall**.
16. **PCA** reduces dimensionality while preserving maximum variance.
17. Knowledge representation uses **visualization and dashboards**.
18. **EDA** is performed before modeling to understand data structure and quality.
19. Histograms show **data distribution**, and box plots detect **outliers**.
20. Bar charts are used for **categorical data**, while histograms represent **continuous data**.
21. Data Mining transforms massive raw data into meaningful knowledge for intelligent decision-making**.**

--Practice questions--

A. Multiple choice questions(mcqs)

1. Which statement best differentiates *Data Mining* from *Machine Learning*?
A. Data Mining focuses only on prediction
B. Machine Learning focuses on storage
C. Data Mining emphasizes discovery of hidden patterns
D. Machine Learning ignores accuracy
Answer: *C*

2. In the KDD process, removing attributes with zero variance is primarily associated with:
A. Data Cleaning
B. Data Selection
C. Data Mining
D. Pattern Evaluation
Answer: *B*

3. If Pearson's correlation coefficient between two variables is **r = 1**, the correct action during preprocessing is to:
A. Normalize both variables
B. Remove both variables
C. Remove one variable to avoid multicollinearity
D. Apply PCA immediately
Answer: *C*

4. Which CRISP-DM phase consumes nearly **80% of total project time** in real-world scenarios?
A. Modeling
B. Business Understanding
C. Data Preparation
D. Deployment
Answer: *C*

5. A Z-score value of **|z| > 3** statistically indicates:
A. Missing value
B. Redundant attribute
C. Outlier
D. Normal data point
Answer: *C*

6. Which visualization technique is most effective for identifying **multicollinearity**?
A. Histogram
B. Box Plot
C. Heat Map
D. Bar Chart
Answer: *C*

7. The Gini Index is primarily used in:
A. Clustering
B. Classification
C. Association Rule Mining
D. Dimensionality Reduction
Answer: *B*

8. In association rule mining, **Confidence** measures:
A. Overall frequency of items
B. Strength of implication
C. Distance between data points
D. Variance of attributes
Answer: *B*

9. Which metric answers the question *"How many relevant instances were correctly identified?"*
A. Precision
B. Accuracy
C. Recall
D. Support
Answer: *C*

10. PCA reduces dimensionality by projecting data onto directions of:
A. Minimum variance
B. Random vectors
C. Maximum variance
D. Zero correlation
Answer: *C*

B. Short Answer Questions (SAQ)

1. Define **Data Mining** and state its primary goal.
2. Explain the **Data → Information → Knowledge → Decision** flow with one example.
3. Why is Data Mining described as a solution for being *"data rich but information poor"*?
4. What is the difference between **Statistics** and **Data Mining**?
5. Define **Information Gain** and its role in feature selection.
6. What is **Data Preprocessing**, and why is it critical in KDD?
7. Explain the **1.5 × IQR rule** for outlier detection.
8. What does a **correlation coefficient (r = 0)** signify?
9. List any two applications of Data Mining in healthcare.
10. Differentiate between **Bar Chart** and **Histogram**.

C. Long Questions (LQ)

1. Explain **Data Mining** in detail. Discuss its goals, importance, and real-world applications.
2. Describe the **Knowledge Discovery in Databases (KDD)** process with a neat diagram and explain each step mathematically where applicable.
3. Compare **Data Mining, Statistics, Machine Learning, and Database Systems** with suitable analogies and examples.
4. Discuss the **CRISP-DM model** in detail and compare it with the KDD process.
5. Explain **Data Preprocessing** techniques including Data Cleaning, Integration, and Transformation with formulas.
6. Describe **Exploratory Data Analysis (EDA)**. Explain its objectives, steps, and importance before modeling.
7. Explain **Box Plot anatomy** and demonstrate how it is used to detect outliers using the 1.5 IQR rule.
8. Describe **Scatter Plots** and explain the difference between **Correlation (R)** and **Coefficient of Determination (R^2)**.
9. Explain **Pattern Evaluation metrics** such as Accuracy, Precision, Recall, Support, and Confidence with formulas.

10. Explain **Principal Component Analysis (PCA)** with mathematical formulation and its role in knowledge representation.